ISO 9001 readiness assessment

Foundry ERP — module by module comparison against ISO 9001:2015, with IATF 16949 and foundry practice noted where relevant. Assessed 18 October 2026 against the committed code: 129 models, 409 routes, 148 migrations.

Strong in place and used **Good** in place, minor gaps **Partial** some of it exists **Missing** nothing exists **Built, unused** schema only, no screens

1. Modules as they stand

Section	ISO clauses	Status	What is in place	What is missing
Production Products, patterns, pattern ownership, casting analysis, production orders, moulding runs, pours, work lists, teams and stations, sand systems, mould readiness	8.1, 8.5.1, 8.5.2	**Strong**	Pattern revisions with cavity maps and layouts; pattern ownership, custody and amortisation; mould configurations and assemblies; moulding work lists; process controls enforce treatment and cooling windows.	No link from a moulding run to an approved process control plan (the plan tables exist but have no screens).
Melting & Materials Melting heats, furnaces, materials, charge recipes, stock	8.5.1, 8.5.2, 7.1.3	**Strong**	Heat chemistry readings, treatments, temperature, power and process readings; charge batches and posting; material consumption per heat; stock movements, reconciliation and reports; furnace eligibility rules.	Material lots hold the supplier name and certificate only as free text. No incoming inspection record against a plan.
Sales & Planning Customers, enquiries and quotations, sales orders, melting plan	8.2.2, 8.2.3, 8.2.4	**Good**	Enquiry to quotation to order chain with a request wizard; contract reviews with a verdict and separate sales and quality sign-offs; quotation rate settings; melting plan and heat planner.	No record of a change to an order after review (8.2.4 amendments) beyond the audit log. No customer-supplied requirement checklist per order line.
Quality & Dispatch Production queue, casting lots, finished goods, inspections, nonconformance, packing, dispatch, yield and scrap, defect list	8.5.2, 8.6, 8.7, 10.2	**Strong**	Inspections record characteristic, method, limits, measured value, equipment and tester; NCRs carry containment, root cause, corrective and preventive action, disposition, owner, target date, closure and verification; quality gates block pour and lot release; casting certificate pulls chemistry and thermal results; yield and scrap reporting.	Corrective action lives only inside an NCR or process event, so there is no single CAPA register and no separate effectiveness check. Inspections consume approved inspection plans but no screen can create one.
Metallurgy Iron grades, thermal tests, sand lab	8.5.1, 9.1.1	**Strong**	Characteristic limits per iron grade, flagged as required for release; thermal analysis with a fitted calibration model and curve analysis; sand systems, recipes and sand tests with base sand data.	No trending or capability analysis across heats; results are judged one at a time.
Equipment & Tooling Flasks, maintenance dashboard and records, tooling life	7.1.3, 7.1.5	**Partial**	Maintenance plans with day and meter triggers; breakdowns; meter readings; equipment capabilities and work centre assignments; tooling life limits with a dashboard; maintenance history and insights; maintenance files now on the shared attachment system.	No calibration anywhere: the equipment table has no calibration interval, due date, certificate or status, and there is no gauge or instrument register. Inspections and process measurements keep the instrument as free text, so a measurement cannot be traced to a calibrated device.

Section	ISO clauses	Status	What is in place	What is missing
People (HR) Employees, attendance and overtime, roster and shifts, leave, payroll, org chart, directory, exits and clearance, HR policies, organisation setup	7.1.2, 7.2, 7.3	Partial	Full employee master with departments, designations, grades, locations and cost centres; attendance, roster and holidays; leave with ledger and approvals; payroll, salaries, advances and settlements; exits and clearance; HR policies with per-version acknowledgement.	No training records, skills matrix, qualification or certification expiry, and no competence evidence per role. Pourers, welders and inspectors normally need these on file.
Settings & governance Audit trail, approvals and policies, users and roles, documents, codes and marking, custom fields, data import, display	5.3, 7.5, 9.1	Strong	Every model change is written to the audit log; approval requests and policies decide which actions need sign-off; role based permissions with per-module file rights; attachments with categories, checksums, optional versioning, cross-record links and pinned versions; generated codes and engraving marks.	No document approval states (draft, approved, obsolete) on attachments, and no document expiry or review date.

2. ISO 9001:2015 clause by clause

Clause	Requirement	Where it lives today	Status	Comment
4.4	QMS and its processes	Production stage definitions, process routes	Partial	Stage definitions exist and are used as quality gates. Process routes and manufacturing methods are schema only.
5.3	Roles, responsibilities, authorities	Roles and permissions, approval policies	Strong	Twenty one permissions across ten seeded roles; approval policies name who decides what.
6.1	Risk and opportunity	—	Missing	No risk register. No FMEA (design or process).
6.2	Quality objectives	Dashboards, yield report	Partial	The numbers exist, but no objective with a target, owner and review date.
7.1.2	People	HR module	Strong	Employees, organisation, roster and attendance.
7.1.3	Infrastructure	Equipment, furnaces, maintenance plans	Strong	Maintenance plans with triggers, breakdowns and meter readings.
7.1.4	Process environment	Sand lab, process controls	Partial	Sand and process conditions are recorded; no general environmental monitoring.
7.1.5	Monitoring and measuring resources	Thermal calibration only	Missing	The single biggest gap. No calibration register, intervals, due dates, certificates or overdue block. Thermal analysis has its own fitted model, which is not a gauge calibration record.
7.2	Competence	—	Missing	No training records, skills matrix or qualification expiry.
7.5	Documented information	Attachments; controlled documents schema	Partial	Attachments now give categories, checksums, permissions, versioning, links and pinned versions. The controlled document tables with revisions and approvals have no screens.
8.2.2 / 8.2.3	Customer requirements and review	Enquiries, quotations, contract reviews	Good	Contract review with verdict and two sign-offs per order.

Clause	Requirement	Where it lives today	Status	Comment
8.3	Design and development	Product specifications schema	**Built, unused**	Specifications, requirements and approvals are designed but have no screens.
8.4	Externally provided processes and products	Material lots (text fields only)	**Missing**	No supplier master, approved vendor list, supplier scoring or incoming inspection. Supplier name and certificate are free text on a material lot.
8.5.1	Production and service provision	Production, melting, moulding, pouring	**Strong**	Orders, runs, pours, heats, readings and process control enforcement.
8.5.2	Identification and traceability	Codes, marks, heats, lots, dispatch	**Strong**	Generated codes and engraved marks; heat to pour to lot to packing to dispatch chain; order trace screen.
8.5.4	Preservation	Packing records, stock	**Partial**	Packing and stock are recorded; no shelf life or storage condition rules.
8.6	Release of products	Quality gates, casting certificate	**Strong**	Pour and lot release gates on required characteristics; certificate pulls chemistry and thermal data.
8.7	Control of nonconforming output	NCRs, defect list, holds	**Strong**	NCR with severity, disposition, quantity affected and closure; lot hold and release.
9.1.2	Customer satisfaction	—	**Missing**	No complaints register and no satisfaction measure.
9.2	Internal audit	—	**Missing**	No audit programme, checklists or findings.
9.3	Management review	—	**Missing**	No review record with inputs and outputs.
10.2	Nonconformity and corrective action	NCR and process control event fields	**Partial**	CAPA fields exist inside NCRs and process events, but there is no cross-source register and no effectiveness verification step.
10.3	Continual improvement	Yield, scrap, insights	**Partial**	Reporting exists; improvement actions are not tracked as records.

3. Foundry and automotive practice

Capability	Status	Comment
Heat and lot traceability	**Strong**	Heat to pour to mould assembly to lot to packing to dispatch, with an order trace screen.
Chemistry per heat	**Strong**	Heat chemistry readings with limits per iron grade, required for release.
Thermal analysis (carbon equivalent, liquidus)	**Strong**	Tests with curve analysis and a fitted calibration model.
Sand testing	**Strong**	Sand systems, recipes, base sand and test records.
Material certificates (EN 10204 3.1)	**Partial**	A certificate reference as text on a material lot; the file can be attached but is not required or validated.

Capability	Status	Comment
Certificate of conformance to customer	**Good**	A printable casting certificate per lot, pulling chemistry, thermal results and inspections.
Pattern and tooling life	**Strong**	Tooling life limits, maintenance plans, cycles, ownership and custody.
Control plan (IATF 16949)	**Built, unused**	Process control plans, items, measurements and events exist in the schema; only the measurement and event parts are reachable.
PPAP package	**Missing**	No PPAP assembly or submission record.
FMEA (design and process)	**Missing**	No FMEA at all.
SPC, Cpk, capability studies	**Missing**	Process measurements hold the data; nothing computes charts or capability.
MSA, gauge R&R	**Missing**	No measurement system analysis.
Layered process audits	**Missing**	No audit module of any kind.
Scrap and yield analysis	**Strong**	Yield and scrap report with a defect list and defect capture on runs and lots.

4. Designed but not reachable

These tables exist in the database with no screens behind them. They are the fastest route to clause 7.5 and 8.3, because the hard modelling is already done.

Table	What it was designed to hold	Reachable today
`controlled_documents`	Document register: number, title, type, owner, status, release and obsolete dates, current revision	None
`controlled_document_revisions`	Revision with file, checksum, status, change reason, prepared and approved by, effective and superseded dates	None
`document_approval_records`	Per-step approval decisions with approver, role and comments	None
`product_specifications`	Specification per product revision, with governing standard, drawing number and revision	Model only
`product_specification_requirements`	Characteristics with limits, method, sampling and criticality	None
`product_specification_approvals`	Approval steps for a specification	None
`inspection_plans / items`	Inspection plan per product revision with characteristics, methods and sampling	Read by the inspection screen; cannot be created
`process_control_plans`	Control plan per product revision, method, route and operation	None
`manufacturing_methods / process_routes`	How a product revision is made, with furnace and sand options	Model only

5. Suggested order of work

Each item should follow the principle already used for attachments: sensible for a small foundry by default, with the heavier requirements switched on in settings by a certified one.

#	Work	Clause	Size	What it involves
1	**Calibration and gauge register**	7.1.5.2	Small	Add calibration interval, due date, status, certificate and responsible person to the existing equipment table; add a gauge register; block or warn when an overdue instrument is used; link the instrument on inspections and process measurements to a real record. Smallest build, largest audit exposure removed.
2	**Supplier master and incoming inspection**	8.4	Medium	A supplier record with approval status and scoring; link material lots to it; an incoming inspection against characteristics; require the mill certificate as an attachment on the lot.
3	**Switch on the controlled document layer**	7.5, 8.3	Medium	The schema is already designed. Add screens for controlled documents, revisions and approvals, and for product specifications and inspection plans. This also delivers the document approval states and review dates still outstanding on attachments.
4	**Training and competence records**	7.2	Medium	Training records, a skills matrix per designation, and certification expiry with reminders. Shares the expiry mechanism with document expiry.
5	**Clause 9 records**	9.1.2, 9.2, 9.3	Medium	Customer complaints, internal audit programme with findings, management review with inputs and outputs, and quality objectives with targets and owners.
6	**CAPA register and effectiveness check**	10.2	Small	One register fed by NCRs, complaints, audit findings and supplier issues, with an effectiveness verification step and a due date.
7	**SPC and capability**	9.1.3	Medium	Control charts and Cpk over the process measurements already collected.

Assessment based on the code committed to the master branch at 18 October 2026. Row counts in the local development database were used only to confirm which tables are unreachable; they do not indicate production use.